Quality Assessment of the Safety of the Intended Functionality in Driver Assistance Systems

A study of ISO 21448 (SOTIF) with an adaptive cruise control case study

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Note on numerical examples. The hazard ratings, acceptance criteria and perception trade-off curves (Sections 4, 9 and 11) are illustrative values chosen to demonstrate the method. The controller test results in Sections 11.3 to 11.6 come from the author's own simulation project and are not data from a production vehicle. Requirements of the standards are cited to the published sources in the reference list.

Abstract

Driver assistance systems such as adaptive cruise control and automatic emergency braking can cause harm even when no component has failed. A radar that cannot separate a stopped car from an overhead sign, or a controller that was never shown a certain traffic situation, behaves exactly as designed and is still unsafe. ISO 26262 does not address this kind of risk. This article examines how the quality of such systems is assessed under ISO 21448:2022, Safety of the Intended Functionality (SOTIF). It follows the SOTIF activities from specification and hazard evaluation, through the analysis of functional insufficiencies and triggering conditions, functional modifications and the four scenario areas, to the verification and validation strategy, validation targets and the release decision. Particular attention is given to the statistics of validation, to scenario-based testing and to quality metrics for perception and decision making. A case study of an adaptive cruise control system, including results from the author's own reinforcement-learning ACC simulation, shows how triggering conditions are discovered, how a functional modification removes them, and why zero collisions in a test campaign is evidence but not proof. The article ends with the open challenges: unknown scenarios, machine learning, regulation of automated driving and the credibility of simulation.

Keywords: ISO 21448, SOTIF, driver assistance, adaptive cruise control, functional insufficiency, triggering condition, scenario-based testing, validation target, reinforcement learning.

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List of abbreviations

ACC	Adaptive cruise control	ADAS	Advanced driver assistance system
AEB	Automatic emergency braking	DQN	Deep Q-network
HIL	Hardware-in-the-loop	ODD	Operational design domain
PD	Proportional-derivative (controller)	RL	Reinforcement learning
SOTIF	Safety of the intended functionality	TTC	Time to collision
STPA	System-theoretic process analysis	V&V	Verification and validation

1. Introduction

1.1 Driver assistance in modern vehicles

Driver assistance has moved from a luxury option to a standard feature. Adaptive cruise control (ACC) holds a speed set by the driver and, when a slower vehicle appears ahead, follows it at a chosen time gap. Automatic emergency braking (AEB) brakes for an obstacle the driver has not reacted to. Lane keeping systems steer the vehicle back into its lane. All of them share one structure: sensors observe the environment, algorithms interpret what they see and decide what to do, and actuators carry out the decision [5].

The difficulty lies in the middle step. A brake pedal has a fixed, fully specified job. A perception and decision chain must deal with an open world: endless combinations of road layouts, weather, light, other road users and their behaviour. No specification can list them all, and no sensor sees perfectly. A system can therefore be free of faults, built exactly to its specification, and still do the wrong thing in a situation its designers did not foresee.

1.2 Why a second safety standard is needed

ISO 26262 [2] controls the risk from *malfunction*: random hardware failures and systematic faults in hardware and software. It asks whether the system does what it was designed to do. For a perception-based function that question is necessary but not sufficient, because the design itself may be insufficient. A well-known example is a radar-based ACC that ignores stationary objects to avoid braking for bridges and road signs, and therefore does not react to a stopped car at the end of a traffic queue. Nothing has failed. The function has a limitation.

ISO 21448 [1], Safety of the Intended Functionality, addresses this gap. In this article, *quality assessment* means the systematic evaluation of the evidence that the residual risk caused by the intended functionality, including its known limitations and reasonably foreseeable misuse, is acceptably low. Where ISO 26262 relies on failure rates and process rigour, SOTIF relies on scenario analysis, statistical validation and a continuous search for situations that nobody has yet thought of.

1.3 Objectives and structure

The objectives of this article are to explain the concepts and activities of ISO 21448, to show how hazards caused by functional insufficiencies are identified and evaluated, to describe the techniques used to verify and validate a perception-based function, and to apply them to a realistic example. Section 2 introduces the standard and its relation to ISO 26262. Section 3 presents the SOTIF activities. Section 4 covers hazard identification and evaluation. Section 5 explains functional insufficiencies and triggering conditions, and Section 6 the four scenario areas. Section 7 describes functional modifications. Section 8 covers the verification and validation strategy and validation targets, and Section 9 quality metrics for perception and decision. Section 10 deals with release and the operation phase. Section 11 is the case study of adaptive cruise control, Section 12 discusses current challenges, and Section 13 concludes.

2. SOTIF and ISO 21448

2.1 The concept of the intended functionality

ISO 21448 defines SOTIF as the absence of unreasonable risk due to hazards resulting from functional insufficiencies of the intended functionality or its implementation, or from reasonably foreseeable misuse by persons [1]. Two causes are named in this definition.

A **functional insufficiency** is either an *insufficiency of specification*, where the specification itself is incomplete or wrong for some situation, or a *performance insufficiency*, where a component such as a sensor or an algorithm cannot meet its specification in some conditions. **Reasonably foreseeable misuse** is use that

the manufacturer did not intend but that results from predictable human behaviour, for example a driver who relies on ACC to stop for every obstacle, or who stops watching the road.

The third key term is the **triggering condition**: a specific condition of a scenario that activates a functional insufficiency and so leads to hazardous behaviour [1]. Heavy spray from a truck, low sun in the camera, a stationary vehicle under a bridge, or a vehicle cutting in very close are typical examples. Much of the quality work in SOTIF consists of finding triggering conditions before customers do.

2.2 History and scope

Work on SOTIF began in the same ISO working group that maintains ISO 26262, once it became clear that the second edition of ISO 26262 would not cover limitations of the intended function. The first result was ISO/PAS 21448:2019, a publicly available specification aimed at emergency intervention systems and driver assistance at SAE automation levels 1 and 2 [3], [6]. The full international standard ISO 21448:2022 followed in 2022. It widened the scope to all levels of driving automation, while noting that higher levels may need additional measures, and it added guidance on topics such as scenario-based validation and machine learning [1].

2.3 Relationship with ISO 26262 and ISO/SAE 21434

Figure 1 places SOTIF next to the two other pillars of automotive safety and security. All three standards are concerned with hazardous behaviour of the vehicle, but they address different causes. ISO 26262 covers malfunction, ISO 21448 covers the intended function and its misuse, and ISO/SAE 21434 [4] covers malicious attacks. A real product has to satisfy all three, and its safety case must show that no cause has been left without an owner.

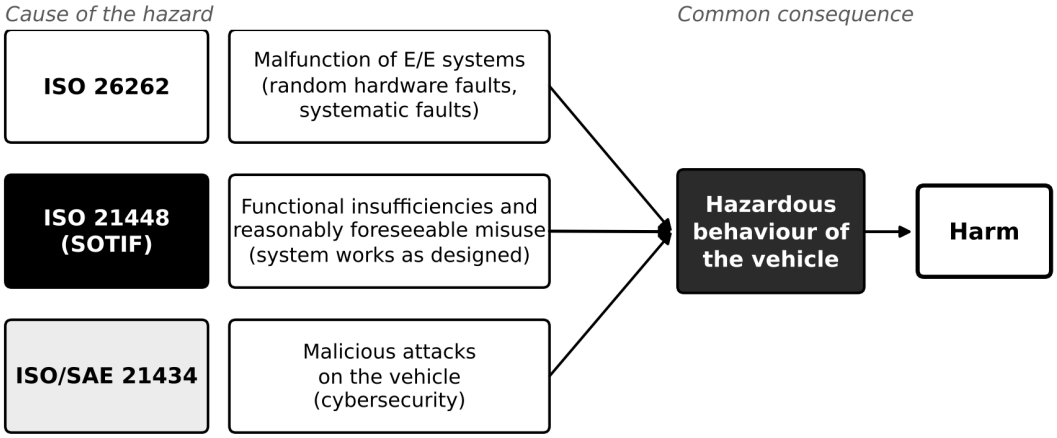


Figure 1. Three standards, three causes of the same hazardous behaviour. SOTIF covers hazards that arise when no component has failed.

Table 1 compares the two safety standards. They are complementary rather than competing. ISO 21448 states that hazards caused by E/E malfunction are handled by ISO 26262 and are outside its scope [1]. In practice the same team usually runs both, and the two hazard analyses share the same list of hazardous behaviours of the vehicle.

Table 1. Comparison of ISO 26262 and ISO 21448 [1], [2].

Aspect	ISO 26262 (functional safety)	ISO 21448 (SOTIF)
Cause addressed	Malfunction: faults and failures of E/E systems	Functional insufficiencies and reasonably foreseeable misuse
Central question	Does the system do what it was designed to do?	Is what it was designed to do safe enough in the real world?
Risk rating	ASIL from severity, exposure and controllability	Severity and controllability, then acceptance criteria and validation targets
Main evidence	FMEDA, hardware metrics, process rigour, code coverage	Scenario analysis, triggering conditions, statistical validation
Quantitative targets	SPFM, LFM, PMHF	Validation targets, e.g. distance or hours without hazardous behaviour
Typical finding	Stuck sensor, bit flip, software bug	Missed object in fog, wrong target on a curve, phantom braking

3. The SOTIF Activities

ISO 21448 organises its activities as an iterative flow rather than a strict V. Figure 2 shows the main clauses. The function is specified (Clause 5), hazards are identified and evaluated (Clause 6), functional insufficiencies and triggering conditions are analysed (Clause 7), and the function is modified to reduce the risk (Clause 8). A verification and validation strategy with validation targets is then defined (Clause 9). The evaluation of known hazardous scenarios (Clause 10) and the search for unknown ones (Clause 11) produce the evidence. Clause 12 judges whether SOTIF has been achieved and recommends release or not, and Clause 13 keeps watching the function once it is on the road [1].

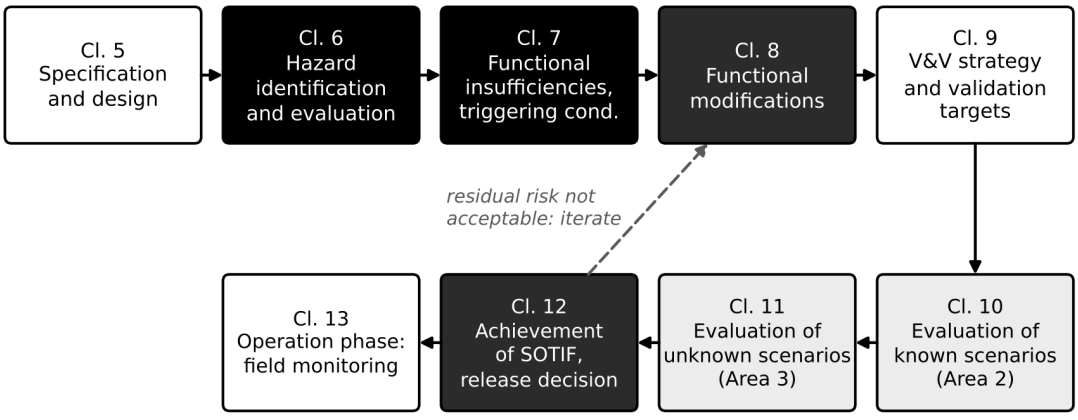


Figure 2. The main SOTIF activities of ISO 21448:2022 and the iteration loop. If the residual risk is not acceptable, the function is modified and evaluated again.

The loop from Clause 12 back to Clause 8 is the heart of the method. If testing reveals a triggering condition with unacceptable risk, the function is changed, for example by improving a sensor, adding a check or restricting where the function may operate, and the evaluation is repeated. Development ends not when all tests pass, but when the evidence shows that the residual risk meets the acceptance criteria.

Table 2 lists the main work products of each activity, together with the quality question that an assessor asks of it. Read from top to bottom, it is a compact picture of what a SOTIF quality assessment actually examines.

Table 2. Main work products of the SOTIF activities and the quality question asked of each.

Activity	Key work products	Quality question
Specification and design (Cl. 5)	Functional description, operational design domain (ODD), system limits, sensor capabilities	Is the intended behaviour, including its limits, described precisely enough to analyse?
Hazard evaluation (Cl. 6)	Hazardous behaviours, severity and controllability, acceptance criteria	Are all relevant hazards found, and is each acceptance criterion justified?
Insufficiencies (Cl. 7)	Functional insufficiencies, triggering conditions, misuse analysis	Were enough methods and sources used for the list to be credible?
Modifications (Cl. 8)	Changed specification or design, ODD restrictions, driver warnings	Is each modification linked to a named triggering condition?
V&V strategy (Cl. 9)	Validation targets, test plan, scenario catalogue	Are targets derived from the acceptance criteria, and are they achievable?
Evaluation (Cl. 10, 11)	Test, simulation and field reports	Is the evidence sufficient and independent of the data used in development?
Release and operation (Cl. 12, 13)	SOTIF release argument, field monitoring plan	Is residual risk acceptable, and will new hazards be detected after release?

4. Hazard Identification and Evaluation

4.1 From function to hazardous behaviour

Clause 6 starts from the functional description and asks which behaviours of the vehicle, caused by the intended function, could lead to harm. For a longitudinal function the list is short but important: braking when it should not (unintended or excessive deceleration), not braking enough when it should (insufficient deceleration), and accelerating when it should not. Each hazardous behaviour is combined with the operational situations in which it matters, much as in an ISO 26262 hazard analysis, and the two analyses are often carried out together [1], [2].

4.2 Severity and controllability

Each resulting hazardous event is evaluated with the severity and controllability classes known from ISO 26262, summarised in Table 3. Exposure is not rated as a class in the same way. In SOTIF, the frequency of the triggering situations enters later, through the acceptance criteria and the validation targets. If a hazardous event can cause no injury (S0) or is controllable in general (C0), no further SOTIF activity is needed for it. Otherwise the event must be shown to meet an acceptance criterion [1].

Table 3. Evaluation of hazardous events in ISO 21448 [1], [2].

Parameter	Classes	Use in SOTIF
Severity (S)	S0 to S3	Potential harm of the hazardous event, from no injuries (S0) to life-threatening or fatal injuries (S3). S0 ends the SOTIF analysis for that event.
Controllability (C)	C0 to C3	Whether the driver or other road users can avoid the harm, from controllable in general (C0) to difficult to control or uncontrollable (C3). C0 ends the analysis.
Exposure	Not rated as a class	Taken into account through the frequency of triggering conditions in the acceptance criteria and validation targets.
Acceptance criterion	Quantitative or qualitative	The level of residual risk regarded as not unreasonable, for example a maximum rate of a hazardous behaviour.

4.3 Acceptance criteria

The acceptance criterion is the quantity at which the whole validation effort is aimed. It states how often, at most, a hazardous behaviour may occur. Common ways to derive it are comparison with human driving, where the system must perform at least as well as an attentive driver in the same situations; comparison with systems already accepted in the market; and the principle that risk should be reduced as far as reasonably practicable [1]. An illustrative criterion might read: *unintended deceleration above 3 m/s^2 lasting longer than 0.5 s shall occur less than once in 10^5 hours of operation.*

The quality of this step is judged by its rationale. A target chosen because it is easy to test is not an acceptance criterion. A target derived from accident statistics and stated together with its assumptions is.

5. Functional Insufficiencies and Triggering Conditions

5.1 The cause-effect chain

Figure 3 shows how SOTIF describes the path from a situation on the road to harm. A triggering condition activates a functional insufficiency. The insufficiency produces an output insufficiency, such as a missed object, a wrong target or a ghost object that does not exist. If nothing downstream catches it, the output leads to hazardous behaviour of the vehicle, and harm follows only if the situation is not controlled by the driver or other road users. Reasonably foreseeable misuse acts late in the chain: a driver who does not watch the road removes the controllability on which the design relies.

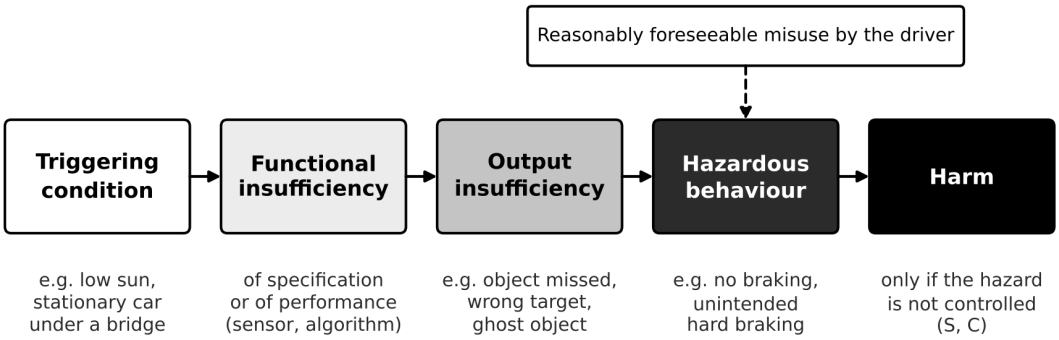


Figure 3. The SOTIF cause-effect chain from triggering condition to harm, with ACC examples under each step.

The chain also shows where quality can be improved. A triggering condition can be avoided by restricting the operational design domain; the insufficiency can be removed with a better sensor or algorithm; the output insufficiency can be caught by a plausibility check or a second sensor; and controllability can be kept by warning the driver in good time.

5.2 Analysis methods

Clause 7 asks for a systematic search for insufficiencies and triggering conditions. No single method is sufficient, so several are combined:

- **Specification against the ODD.** The operational design domain is the set of conditions, such as road types, speeds, weather and lighting, in which the function is designed to work. Each condition is checked against what the specification says the function will do.
- **Sensor and algorithm limitation analysis**, using datasheets and test data on detection range, resolution, field of view and behaviour in rain, fog, glare or clutter.

- **Guide-word analysis and STPA.** System-theoretic process analysis [15] is well suited to finding unsafe control actions and unsafe interaction between the function and the driver.
- **Lessons learned** from previous products, accident databases and field returns.
- **Exploratory testing** in simulation and on the road, which is the main source of triggering conditions that no analyst foresaw.

5.3 Typical triggering conditions for ACC

Table 4 lists triggering conditions that are widely discussed for radar and camera based longitudinal functions. Each row names the insufficiency it activates and the hazardous behaviour that may follow. A list like this is only a starting point. The quality question for the assessor is not whether the list exists, but whether it was built with enough methods and sources to be credible, and whether each entry is linked either to a modification or to an argument that its residual risk is acceptable.

Table 4. Typical triggering conditions for a radar and camera based ACC function.

Triggering condition	Insufficiency activated	Possible hazardous behaviour
Stationary vehicle at the end of a queue, especially under a bridge or sign gantry	Radar filtering of stationary targets used to suppress clutter	No braking or late braking
Metal plate or drain cover on the road, overhead sign on a crest	Strong radar reflection classified as an in-path obstacle	Unintended braking (phantom braking)
Vehicle in the adjacent lane while driving through a curve	Target selection relies on a path predicted from yaw rate	Braking for the wrong vehicle, or ignoring the right one
Close cut-in at low relative speed	Object enters the field of view late; tracking needs time to confirm it	Late or harsh braking
Heavy rain, spray, snow on the radome	Reduced radar range or blocked sensor	Late braking or sudden loss of function
Low sun, tunnel exit, glare	Camera saturation	Missed classification, degraded fusion
High-floor trailer or vehicle with an overhanging load	Reflection point differs from the nearest point of the vehicle	Gap estimated as too large
Lead vehicle far ahead brakes hard to a stop	Controller specification ignores lead deceleration	Braking starts too late to stop within the ACC limit

6. The Four Scenario Areas

A distinctive idea of SOTIF is to divide all possible scenarios into four areas according to two questions: is the scenario known to the developers, and is it hazardous with the current design? Figure 4 shows the result [1].

- **Area 1, known and not hazardous:** the function has been shown to behave safely.
- **Area 2, known and hazardous:** limitations that have been found and must be removed by modification or shown to carry acceptable residual risk.
- **Area 3, unknown and hazardous:** the dangerous blind spot of the project, scenarios nobody has thought of yet.
- **Area 4, unknown and not hazardous:** harmless, and of no direct concern.

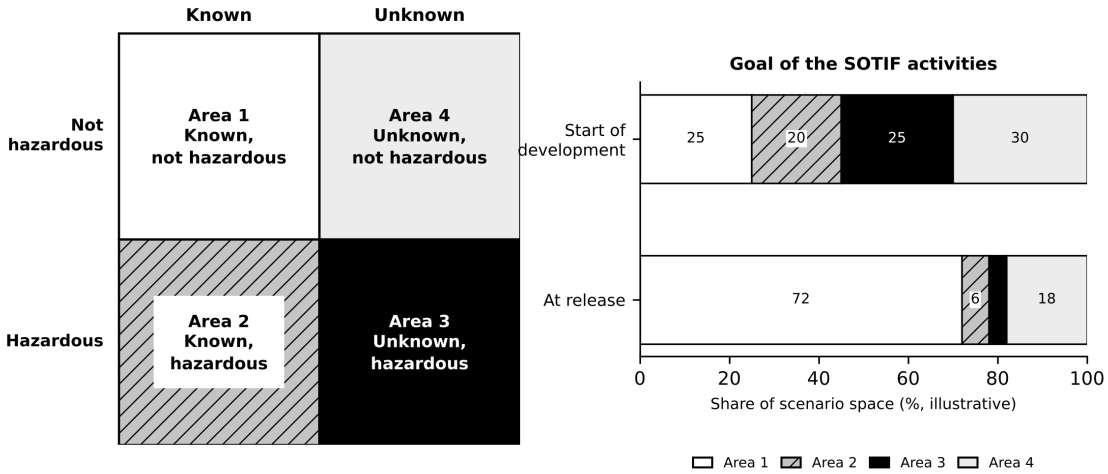


Figure 4. The four scenario areas of SOTIF (left) and the goal of the activities (right): enlarge Area 1 and shrink Areas 2 and 3 to an acceptable level. Proportions are illustrative.

The aim of the SOTIF activities is to move scenarios out of Areas 2 and 3. Analysis and testing move scenarios from the unknown side to the known side, turning Area 3 into Area 2 and Area 4 into Area 1. Functional modifications then move scenarios from Area 2 into Area 1. Clause 10 evaluates Area 2 with targeted tests of known scenarios. Clause 11 estimates the residual risk of Area 3 through broad exploration, random and long-duration testing and field data, because, by definition, the unknown cannot be tested one scenario at a time.

This picture changes how quality is measured. A test campaign that only replays known scenarios can show that Area 2 is under control, but it says almost nothing about Area 3. An honest quality assessment therefore asks two separate questions: how well are the known hazards handled, and how much effort has gone into finding the ones that are still unknown?

7. Functional Modifications

When a triggering condition leads to unacceptable risk, Clause 8 requires a functional modification. Modifications act at different points of the chain in Figure 3. Table 5 summarises the main types, with ACC examples [1], [5].

Table 5. Types of functional modification and ACC examples.

Type of modification	Effect on the chain	ACC example
Improve the system	Removes or reduces a performance insufficiency	Higher-resolution radar; camera added for object classification
Add diversity and plausibility checks	Catches output insufficiencies before they act	Brake for a stationary target only when radar and camera agree
Change the specification	Removes an insufficiency of specification	Controller uses lead vehicle deceleration, not only gap and relative speed
Restrict the ODD or the authority	Avoids the triggering condition or limits its effect	Limit deceleration and jerk; switch off in heavy rain or above a set speed
Hand over to the driver	Preserves controllability	Take-over warning when the braking demand exceeds the ACC limit
Reduce misuse	Prevents over-reliance on the function	Clear HMI showing the target vehicle, owner's manual, driver monitoring

Modifications have side effects, and the assessor must check them. Restricting the ODD lowers risk but also lowers usefulness, and a function that switches off too often teaches drivers to ignore it. A plausibility check that suppresses phantom braking can also suppress a braking manoeuvre that was needed. Every

modification must therefore be traced to the triggering condition it addresses and re-evaluated against all hazardous behaviours, not only the one that motivated it.

The performance standard for ACC, ISO 15622 [5], already builds in modifications of the restricting kind. It bounds the automatic deceleration and jerk that the system may command, which keeps ACC a comfort function and leaves emergencies to the driver and to a separate AEB system.

8. Verification and Validation Strategy

8.1 Verification of known scenarios

Clause 10 evaluates Area 2. For every known triggering condition there should be tests that reproduce it and show either that the modification works or that the residual risk is acceptable. Verification takes place at several levels: sensors are tested against their specification in rain chambers and on target ranges, perception algorithms against labelled data, and the complete function in closed-loop scenarios. Scenario catalogues are usually written in a formal description language such as ASAM OpenSCENARIO [13], which makes the same scenario repeatable in simulation, on a hardware-in-the-loop bench and on the proving ground.

8.2 Scenario-based testing

Because the space of scenarios is infinite, testing is organised around scenarios rather than kilometres. A *functional* scenario such as "lead vehicle brakes on a straight motorway" becomes a *logical* scenario with parameter ranges for speeds, gaps and deceleration, and finally many *concrete* scenarios with fixed values. Methods developed in research programmes such as PEGASUS [14] and standardised in ISO 34502 [7] use this hierarchy to search systematically for critical combinations. Figure 5 shows how the test levels share the work.

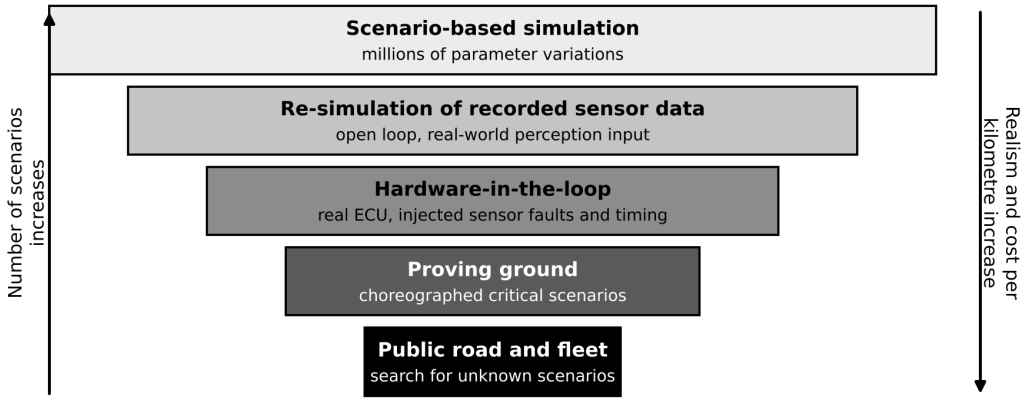


Figure 5. Test levels for SOTIF validation. Simulation covers the most scenarios at the lowest cost; road and fleet driving add realism and the search for unknown scenarios.

Simulation is only as good as its models. A sensor model that never produces a ghost target cannot reveal phantom braking. The credibility of each simulation tool must therefore be argued in its own right, and simulation results that count as validation evidence must be checked against real measurements.

8.3 Validation targets and the statistics of driving

Clause 11 deals with the residual risk of unknown scenarios, and here the main tool is statistics. If a hazardous event occurs randomly at a constant rate λ per kilometre, the probability of driving a distance N without any event is $e^{-\lambda N}$. To show with confidence C that the true rate is below a target λ, the vehicle must drive the following distance without a single hazardous event [10]:

$$N = -\ln(1 - C) / \lambda$$

$$N \approx 3 / \lambda \text{ for } C = 95\ \%$$

The factor of about three at 95 % confidence is known as the **rule of three**. Figure 6 plots the relation. Kalra and Paddock showed that demonstrating a fatality rate equal to that of human drivers in the United States, about 1.09 per 100 million miles, would require some 275 million failure-free miles, roughly 440 million kilometres [9]. Even a large test fleet cannot drive that distance before a product is released.

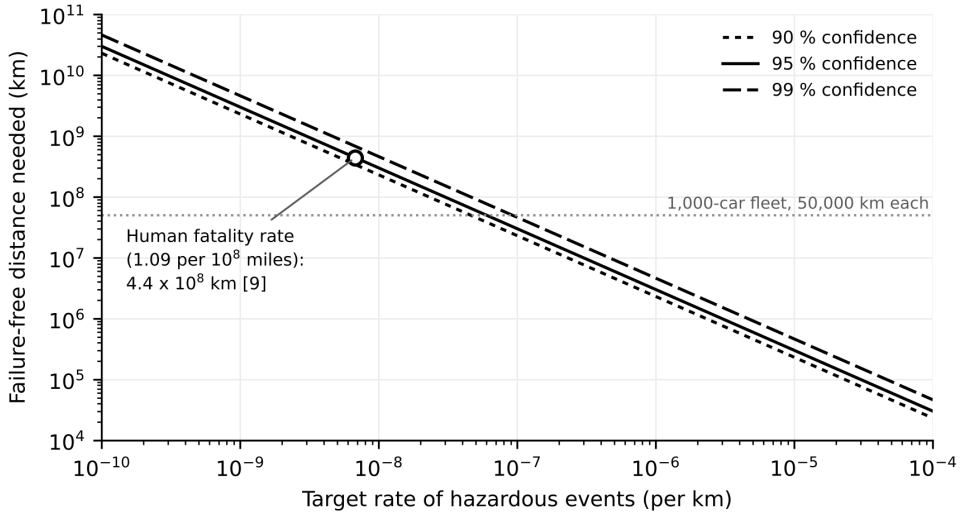


Figure 6. Failure-free distance needed to demonstrate a target rate of hazardous events, for three confidence levels. The distance grows in inverse proportion to the target rate.

This is why SOTIF validation targets are usually set per hazardous behaviour and per triggering situation, not for "safety" as a whole, and why they are reached by combining sources: re-simulation of recorded sensor data, which replays millions of kilometres of real perception input; simulation with validated models; targeted proving-ground tests; and long-duration road driving. The quality question is whether these sources are combined honestly. Kilometres driven in a simulator that cannot produce a given triggering condition do not count towards the target for that condition [1], [8].

9. Quality Metrics for Perception and Decision

9.1 The trade-off between false positives and false negatives

A perception-based function makes two kinds of mistakes. A **false negative** misses a real object and can lead to a collision. A **false positive** reacts to an object that is not there, or not in the path, and leads to unintended braking, which can cause a rear-end collision and erodes the driver's trust. Both are hazardous behaviours, and they pull in opposite directions: a lower detection threshold reduces misses but increases phantom braking. Figure 7 illustrates the trade-off and how sensor fusion improves it.

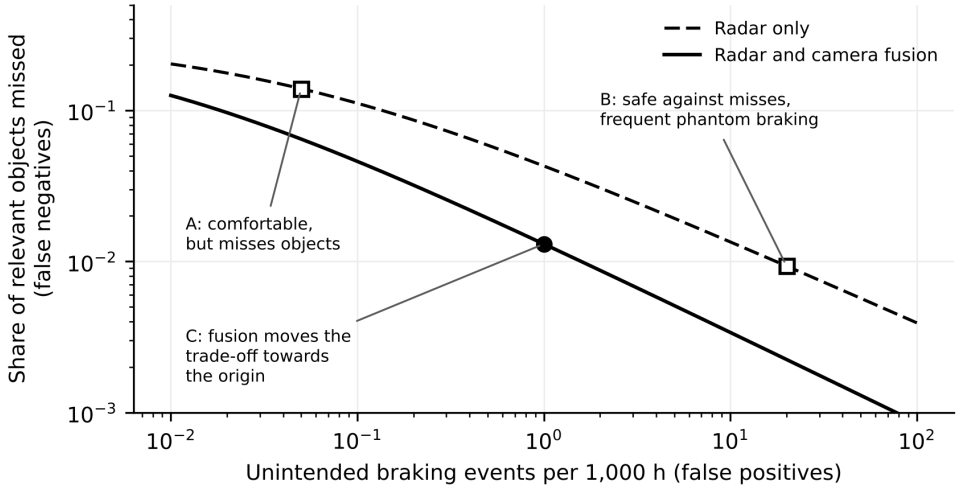


Figure 7. Illustrative trade-off between unintended braking and missed objects. Each curve is one sensor configuration, and moving along it changes the detection threshold. Fusion shifts the whole curve.

SOTIF quality assessment therefore never accepts a single accuracy figure. It asks for both error types, each measured in the unit that matters for the hazard, such as missed objects per relevant encounter and unintended braking events per hour, and it asks for the chosen operating point to be justified against the acceptance criteria of both hazardous behaviours.

9.2 Metrics used in assessment

Table 6 collects the metrics most often used as SOTIF evidence. None of them is sufficient alone. Together they describe how well the function perceives, how close it came to harm in the scenarios that were tested, and how representative those scenarios were.

Table 6. Quality metrics for perception-based driver assistance.

Metric	What it shows	Typical use
Precision and recall of object detection	Share of detections that are real; share of real objects that are detected	Perception verification against labelled data
Unintended braking events per hour or per km	Frequency of false-positive hazardous behaviour	Validation target for phantom braking
Minimum time to collision (TTC)	How close the vehicle came to a collision in a scenario	Criticality of closed-loop scenarios
Maximum braking demand	Deceleration that would have been needed to avoid a collision	Whether a situation stayed within the ACC braking limit
Scenario and parameter coverage	Share of the logical scenario space that has been tested	Completeness of known-scenario evaluation
ODD coverage	Share of ODD conditions (weather, road, light) present in the test data	Representativeness of validation
Failure-free distance or time	Exposure accumulated without a hazardous event	Statistical argument for unknown scenarios

For learning-based components there is one more question: are the test data independent of the data used to develop the component? If a model is adjusted until it passes a test set, the test set has become development data and its result is optimistic. Separating training, validation and test data, and recording how often the test set was used, are simple but essential quality measures [11].

10. Release and the Operation Phase

10.1 Evaluating the achievement of SOTIF

Clause 12 brings the evidence together. The assessor checks that all hazardous behaviours have been identified, that known triggering conditions have been addressed, that the validation targets have been met with sufficient confidence, and that the residual risk from unknown scenarios has been argued credibly. The outcome is a recommendation on release. It may be conditional, for example limited to a smaller ODD or tied to further field monitoring [1]. As with ISO 26262, the people who evaluate the evidence should be independent of those who produced it, so that a team does not mark its own work.

10.2 Operation phase activities

Release does not end SOTIF. Clause 13 requires the manufacturer to monitor the function in the field, because a fleet will meet scenarios that development never did. Customer complaints about phantom braking, data recorded around AEB interventions and accident reports are all sources of new triggering conditions. A newly found triggering condition moves a scenario from Area 3 to Area 2 and starts the loop of Figure 2 again, now possibly ending in a software update delivered over the air [1], [4].

11. Case Study: Adaptive Cruise Control

This section applies the method to an ACC function. It combines two kinds of material. The hazard ratings and acceptance criteria are **illustrative**, chosen to show the method. The test results in Sections 11.3 to 11.6 are **real**: they come from the author's own B.Tech project, an ACC controller trained with deep reinforcement learning and compared with a conventional controller in simulation [19]. That project was not developed to ISO 21448, but its testing history is a small, genuine example of the SOTIF loop.

11.1 Function and architecture

Figure 8 shows a typical ACC architecture. Radar and camera feed a fusion and tracking stage. Target selection decides which vehicle is in the ego path, using a path predicted from speed and yaw rate. The longitudinal controller computes an acceleration request; in a conventional system this is a control law, and in the case-study project it is a learned policy. An independent AEB function can request stronger braking, and an arbitration stage passes on the lowest acceleration request. The driver can override at any time, since ACC is a level 1 assistance function [6].

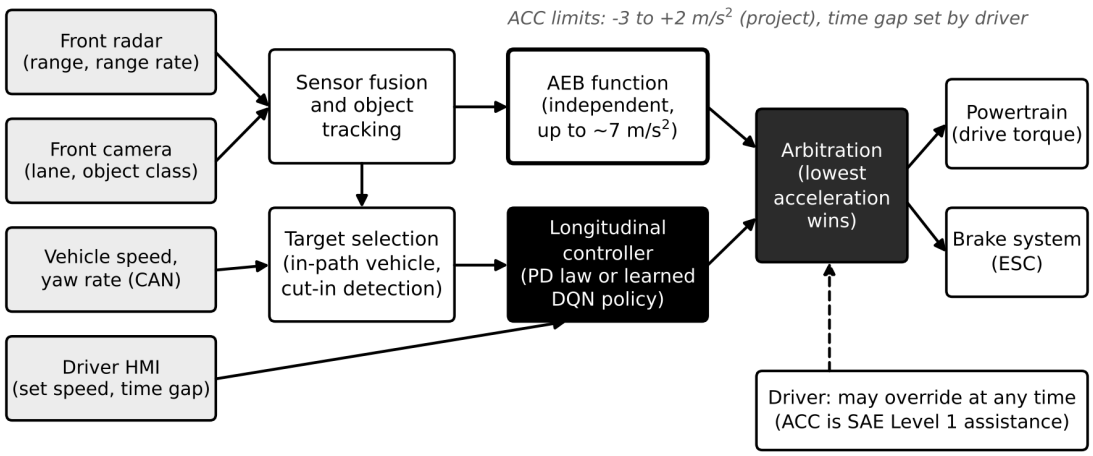


Figure 8. Simplified ACC architecture. The arbitration stage lets the independent AEB function override the comfort controller, whether that controller is a PD law or a learned policy.

In the project, the controller observes six normalised inputs: ego speed, relative speed, gap, the gap error relative to a safe distance $d_{\text{safe}} = 5\ \text{m} + 1.5\ \text{s} \cdot v$, ego acceleration and lead vehicle acceleration. It chooses from ten accelerations between −3 and +2 m/s^2. The policy is a double deep Q-network [16], [17] with two hidden layers of 64 neurons, written from scratch and trained over 800 simulated episodes. The baseline is a constant time-gap PD controller with saturated spacing error and lead-acceleration feed-forward, tuned by grid search with the same reward [19].

11.2 Hazard evaluation

Table 7 shows an illustrative evaluation of four hazardous events. Insufficient deceleration towards a stopped vehicle at motorway speed is the most severe. Phantom braking is rated lower, but it is important because it pulls the design in the opposite direction, which is the trade-off of Figure 7.

Table 7. Illustrative hazard evaluation for the ACC example.

Hazardous event	S	C	Acceptance criterion (illustrative)
Insufficient deceleration when approaching a stopped or hard-braking vehicle at motorway speed	S3	C2	Fewer than 1 event per 10^7 km
Unintended deceleration above 3 m/s^2 with no relevant object (phantom braking) in dense traffic	S2	C2	Fewer than 1 event per 10^5 h
Following the wrong target on a curve while closing on the real lead vehicle	S2	C2	Fewer than 1 event per 10^6 km
Late reaction to a close cut-in at low relative speed	S1	C1	Driver warned within 0.5 s once braking demand exceeds the ACC limit

11.3 Triggering conditions found during the project

The most instructive part of the case study is how triggering conditions were found. Table 8 lists three from the project history. None of them came from a desk analysis. Each was found by a test that exposed the controller to something new, and each was removed by a modification of a type listed in Table 5 [19].

Table 8. Triggering conditions found by testing in the author's ACC project [19].

Found by	Triggering condition	Insufficiency	Modification
Mock-test drives (runs 2 to 3)	Lead vehicle slows while the gap is still large	Reward did not penalise braking that would be needed later, so the controller braked late	"Braking needed" penalty added; baseline spacing error saturated
Development set (run 4)	Lead vehicle slightly faster than the ego vehicle	Reward loophole: tailgating a faster car scored well	Reward corrected (change of specification)
Live demo crash, then a 60-drive stress test (runs 8 to 9)	Lead vehicle far ahead brakes hard to a stop	Specification: the five inputs did not include lead acceleration, so a braking car far ahead looked like a cruising one	Sixth input, lead acceleration; braking-needed term includes the lead's stopping distance

The third case is a textbook SOTIF finding. The controller worked exactly as specified, and the network had been trained correctly. The *specification* of its inputs was insufficient: without lead acceleration the state did not describe the situation completely, so no amount of training could have taught the correct reaction. The fix was a change of specification, not a bug fix. In SOTIF terms, the scenario moved from Area 3 to Area 2 when the live demo crashed, and into Area 1 after the modification was made and tested again.

11.4 Results

Figure 9 and Table 9 summarise the evidence. In the hard-stop stress test, with the AEB safety net switched off, the earlier AI with five inputs crashed in 7 of 60 drives and the PD baseline in 8, while the final AI completed all 60 with a closest gap of 4.7 m. All eight baseline crashes happened when the lead vehicle cruised at 26 or 30 m/s, faster than the 25 m/s set speed, before braking to a stop. The gap had grown large, the saturated spacing error reported no danger, and by the time the closing speed was high, stopping within 3 m/s^2 was no longer possible. This is a triggering condition of the baseline design, found by the same test.

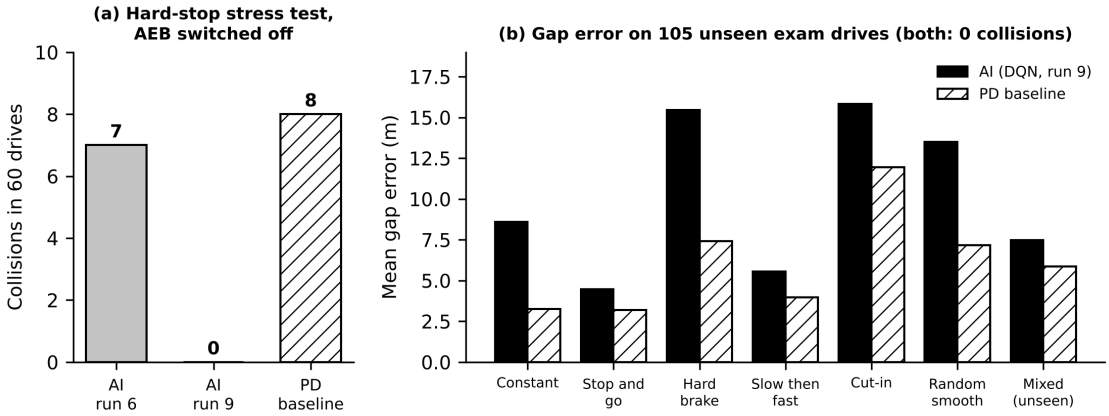


Figure 9. Results from the author's ACC project [19]. (a) Collisions in 60 hard-stop drives with AEB off. (b) Mean gap error per scenario family on the 105-drive exam, in which neither controller collided.

Table 9. Exam results on 105 unseen drives (7 scenario families × 15) and the stress test [19].

Metric	AI (DQN, final)	PD baseline
Collisions in 105 unseen exam drives	0	0
Gap error $\lvert gap - d_{safe} \rvert$ (m)	10.14	6.14
Speed tracking RMSE (m/s)	2.60	2.61
RMS acceleration (m/s^2)	0.85	0.88
RMS jerk (m/s^3)	1.43	0.69
Worst minimum gap (m)	4.85	5.00
Collisions in 60 hard-stop drives, AEB off	0	8

The modification had a price. On average the final AI follows about 9 m further back than the safe distance, so its gap error is worse than the baseline's, and its jerk is about twice as high because it chooses from ten fixed accelerations rather than a continuous value. In SOTIF language, the risk of insufficient deceleration went down, while comfort, and with it driver acceptance, went down too. A quality assessment must report both sides.

11.5 What zero collisions proves

Zero collisions in 105 unseen drives is good news, but it is not a safety proof, and the rule of three shows why. Figure 10 applies it to the project. With no collision in 105 independent drives, the 95 % upper bound on the probability of a collision per drive is about 2.8 %. Counting all 200 crash-free drives of the final model, including the development set and the stress test, it falls only to 1.5 %.

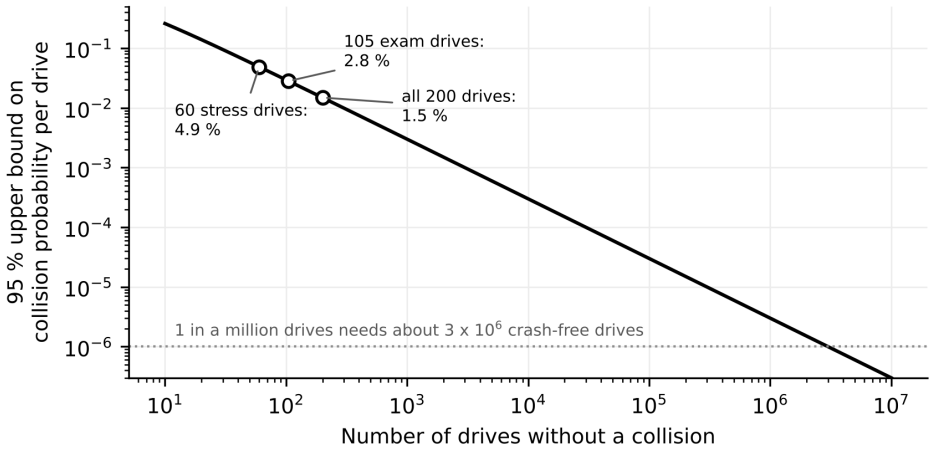


Figure 10. Upper bound (95 % confidence) on the collision probability per drive after a number of drives without a collision, with the project's test campaigns marked.

Each exam drive lasts 60 s, so the 105 drives cover roughly 150 km of simulated driving. Demonstrating the illustrative criterion for insufficient deceleration in Table 7 at 95 % confidence would need about 3×10^7 failure-free kilometres, some five orders of magnitude more. This does not make the project's tests worthless. Their purpose was to find triggering conditions and to compare designs, and they did both. It shows instead why SOTIF separates the evaluation of known scenarios from the statistical argument for unknown ones, and why a learned controller in a real vehicle would need an independent safety layer, such as the AEB arbitration of Figure 8 or a formal safety shield [18], rather than relying on the policy alone.

11.6 Checking the quality criteria

Table 10 collects the quality criteria of the case study against the evidence. This is the form in which a SOTIF argument is reviewed: each criterion has a target, a result and a pointer to the supporting work product.

Table 10. Quality criteria checked against evidence for the ACC case study.

Criterion	Target	Result	Evidence
Hazardous behaviours identified and rated	All, with rationale	Met (illustrative)	Hazard evaluation, Table 7
Known triggering conditions addressed	Each linked to a modification	3 of 3 addressed	Project history, Table 8
Collisions on unseen exam	0 of 105	0	Exam report [19]
Hard-stop stress test, AEB off	0 of 60	0 (baseline 8)	Stress-test report [19]
Test data independent of development	Test set not used for model selection	Met; separate development set, exam runs logged	Project notes [19]
Comfort (RMS jerk)	Not worse than baseline	Not met (1.43 vs 0.69 m/s^3)	Exam report [19]
Statistical target, insufficient deceleration	< 1 per 10^7 km at 95 %	Not demonstrated (about 150 km)	Would need fleet and re-simulation data
Independent safety layer for learned controller	Present and verified	AEB in demo only, not verified	Open item

Read together, the table gives an honest verdict. The project demonstrates the SOTIF loop at a small scale and removes the triggering conditions it found, but it does not supply a release argument, and it was never meant to.

12. Challenges and Emerging Trends

12.1 The problem of unknown scenarios

Area 3 can be made smaller, but it can never be shown to be empty. Statistical arguments rest on the assumptions that test driving is representative of future use and that rare events are independent. Neither holds perfectly: unusual weather, new vehicle designs on the road and changes in driver behaviour all shift the distribution of scenarios after release. This is why field monitoring is a permanent activity rather than a one-time check.

12.2 Machine learning

Learned perception and control raise SOTIF questions directly. Their specification is partly the training data, so coverage of the ODD by the data becomes a safety requirement. Their behaviour between and beyond the training examples is hard to predict, and structural code coverage says little about them. ISO/PAS 8800:2024 [11] addresses safety and artificial intelligence in road vehicles, building on both ISO 26262 and ISO 21448, with guidance on data quality, model verification and monitoring in operation. The case study shows the same lesson on a small scale: the most important fix concerned what the network could see, not how it learned.

12.3 Higher automation and regulation

At SAE level 3 and above the driver is no longer the fallback, so controllability can no longer be assumed, and the evaluation of unknown scenarios carries much more weight. UN Regulation No. 157 on Automated Lane Keeping Systems [12] was the first binding international regulation for a level 3 function, and it asks for a safety management system and scenario-based testing. Scenario-based evaluation frameworks such as ISO 34502 [7] are likely to become the common language between manufacturers and approval authorities.

12.4 Credibility of simulation

Since most validation evidence will come from simulation, the quality of simulation tools becomes a safety question in itself. Sensor models must reproduce the effects that trigger insufficiencies, such as multipath reflections, rain clutter and glare. A model that is too clean produces evidence that is too optimistic. Correlating simulation with measured data, and agreeing clear rules for which simulated kilometres may count towards a validation target, are active areas of work.

12.5 Limits of the SOTIF approach

The standard provides a framework, not numbers. Acceptance criteria are left to the manufacturer, and deriving them from accident statistics involves judgement. The four areas are a powerful mental model but cannot be measured directly. As with the ASIL in ISO 26262, the value of SOTIF depends on the honesty of the people who apply it, and above all on their willingness to look for the scenarios that would prove them wrong.

13. Conclusion

This article has followed how the quality of perception-based driver assistance is assessed under ISO 21448:2022. SOTIF begins where ISO 26262 stops: with a system that works as designed, but whose design is not good enough for every situation on the road. Hazardous behaviours are evaluated for severity and controllability and given acceptance criteria. Functional insufficiencies and their triggering conditions are searched for through analysis and testing, and functional modifications move scenarios from the hazardous areas into the safe one. Verification of known scenarios and statistical validation of unknown ones then supply the evidence for a release decision, and field monitoring keeps that decision valid.

The main lessons are the following:

- A system can be fault-free and still unsafe. SOTIF asks whether the intended function is good enough, not only whether it works.
- Triggering conditions are found mostly by tests that expose the function to new situations, so exploratory testing is a quality activity, not a formality.
- Every functional modification has side effects. In the case study, safer braking cost following accuracy and comfort.
- Zero failures in a test campaign limits the risk only as far as the rule of three allows. Statistical claims need exposure, and exposure needs a combination of fleet, re-simulation and simulation evidence of known credibility.
- Learned components turn part of the specification into data, which is why independent test data and an independent safety layer matter.

For an engineer in robotics and automation, the lesson reaches well beyond cars. Any robot that perceives an open world, from a warehouse vehicle to a field robot, can be free of faults and still meet a situation its designers never imagined. Asking which scenarios are still unknown, and building the evidence to shrink that set, is the discipline that SOTIF teaches.

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